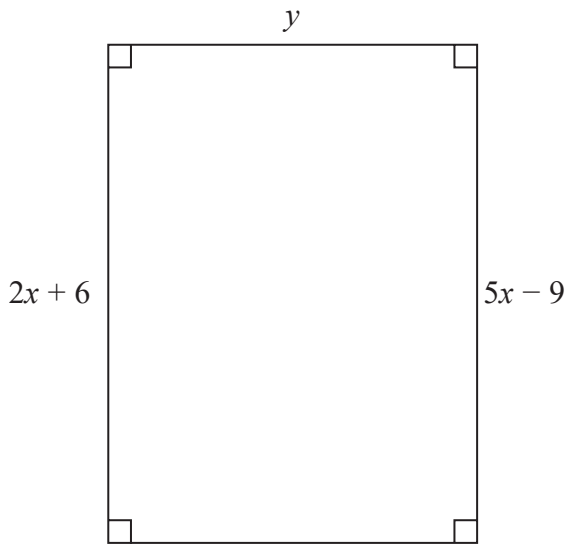
6 Here is a rectangle.



All measurements are in centimetres.

The area of the rectangle is 48 cm^2.

Show that $y = 3$

Opposite sides of a rectangle are equal.

$2x + 6 = 5x - 9$

$-2x \qquad -2x$

$6 = 3x - 9$

1 mark
Form the equation

$+9 \qquad +9$

$15 = 3x$

1 mark
Collect like terms

$\div 3 \qquad \div 3$

$5 = x$

Side length = $2 \times 5 + 6 = 16$ cm.

1 mark
Substitute to find a side

Area = $16y = 48$, so $y = 3$ because $16 \times 3 = 48$.

1 mark
Show that y = 3

(Total for Question 6 is 4 marks)